

Math 121: Commutative Algebra

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Worksheet 1.1: Basics of Rings

Problem 1. Let R be a ring and consider $R[[t]]$ the ring of formal power series with coefficients in R . The goal of this exercise is prove that:

$$(R[[t]])^\times = \{u + tf(t) \mid u \in R^\times, f(t) \in R[[t]]\}.$$

- (a) Elements of $R[[t]]$ can be written as $f = \sum_{k=0}^{\infty} f_k t^k$ for $f_k \in R$. We call f_0 the constant (or lead) term of f . If $f = \sum_{k=0}^{\infty} f_k t^k$ and $g = \sum_{k=0}^{\infty} g_k t^k$ then find a formula for the lead term of fg .
- (b) Using the formula from (a) prove that if f is a unit then $f_0 \in R^\times$.
- (c) Find a formula for the coefficient of t^k appearing in fg .
- (d) Assume that $f_0 \in R^\times$. Prove that f is a unit by explicitly constructing an element $g \in R[[t]]$ such that $fg = 1$. (Hint: the condition that $fg = 1$ places constraints on each coefficient of fg . Use the constraints together with part (c) to recursively solve for the coefficients of g .)

Solution.

- (a) Formula for the constant term of fg is $f_0 g_0$.
- (b) Assume f is a unit. Then there exists some $f' \in R[[t]]$ such that $ff' = 1$. By part (1), the constant term of ff' is $f_0 f'_0$. The constant term of 1 is 1 , so $f_0 f'_0 = 1$. Hence f_0 is a unit, and $f_0 \in R^\times$.
- (c) The formula for the coefficient of t^k is $\sum_{i=0}^k f_i g_{k-i}$.
- (d) Assume $f_0 \in R^\times$. Let its inverse be g_0 . For f to be a unit, every non-leading coefficient of fg must be 0 . That is,

$$fg = f_0 g_0 + \sum_{k=1}^{\infty} \left(\sum_{i=0}^k f_i g_{k-i} \right) t^k$$

where $\sum_{i=0}^k f_i g_{k-i} = 0$ for all k . Then solving for g_k gives us,

$$g_k = -\frac{1}{f_0} \sum_{i=0}^{k-1} f_i g_{k-i} = -g_0 \sum_{i=0}^{k-1} f_i g_{k-i}$$

Problem 2.

- (a) Prove that an element $u \in R$ is a unit if and only if the principal ideal $\langle u \rangle = R$.
- (b) Prove that if ab and b are units, then a is a unit.
- (c) Prove that if u is a unit and $a \in R$, then a and ua generate the same principal ideal.
- (d) In the ring $\mathbb{Z}/\langle 12 \rangle$, determine which elements are units.

Solution.

- (a) ($\implies$) Assume $u \in R$ is a unit.

Then there exists some $u' \in R$ such that $uu' = 1 \in \langle u \rangle$. If 1 is in the ideal generated by u then $\forall r \in R, r \in \langle u \rangle$. Hence $\langle u \rangle = R$.

($\impliedby$) Assume $\langle u \rangle = R$ for some $u \in R$.

Then $1 \in \langle u \rangle$. That is, by definition of an ideal, 1 is a multiple of u . Hence, there exists some $u' \in R$ such that $uu' = 1$. Hence, u is a unit.

- (b) Let $a, b \in R$, and ab, b be units. Then there exist $x, b^{-1} \in R$ such that

$$abx = 1, \quad bb^{-1} = 1$$

Then we have,

$$abx = bb^{-1} \implies \frac{abx}{bb^{-1}} = 1 \implies a \cdot \frac{x}{b^{-1}} = 1$$

Hence, a is a unit and $\frac{x}{b^{-1}}$ is its multiplicative inverse.

- (c) Let $u \in R^\times, a \in R$.

Note that the containment $\langle ua \rangle \subseteq \langle a \rangle$ holds by definition. So we only need to show that $\langle a \rangle \subseteq \langle ua \rangle$. Since u is a unit, there exists some $u^{-1} \in R$ such that $uu^{-1} = 1$. Then, note that $uu^{-1}a = a \in \langle ua \rangle$, i.e. $\langle a \rangle \subseteq \langle ua \rangle$.

Hence a and ua generate the same principal ideal.

- (d) $(\mathbb{Z}/\langle 12 \rangle)^\times = \{1, 5, 7, 11\}$

Problem 3. Given two rings $\mathbf{R}$ and $\mathbf{S}$ we let $\text{Hom}_{\text{Ring}}(\mathbf{R}, \mathbf{S})$ be the set of ring homomorphism from $\mathbf{R}$ to $\mathbf{S}$. When there is little room for confusion we will often drop the subscript Ring and simply write $\text{Hom}(\mathbf{R}, \mathbf{S})$.

- (a) Prove or disprove: If $\mathbf{R}$ and $\mathbf{S}$ are rings, then $\text{Hom}(\mathbf{R}, \mathbf{S})$ is a ring under point-wise multiplication and point-wise addition.
- (b) Give an explicit description of the hom-sets below:
 - (a) $\text{Hom}(\mathbb{Z}, \mathbb{Z})$
 - (b) $\text{Hom}(\mathbb{Z}/\langle 2 \rangle, \mathbb{Z})$
 - (c) $\text{Hom}(\mathbb{Z}, \mathbb{Z}/\langle 2 \rangle)$
 - (d) $\text{Hom}(\mathbb{Z}, \mathbb{C}[x])$
- (c) Based on part (b) state and prove a characterization of $\text{Hom}(\mathbb{Z}, \mathbf{R})$ for any ring $\mathbf{R}$.

Solution.

- (a) Let $f, g \in \text{Hom}(\mathbf{R}, \mathbf{S})$. Note that since ring homomorphisms are unital $f(1_{\mathbf{R}}) = 1_{\mathbf{S}}$ and $g(1_{\mathbf{R}}) = 1_{\mathbf{S}}$. Assume for the sake of contradiction that $\text{Hom}(\mathbf{R}, \mathbf{S})$ is a ring under pointwise addition. Then,

$$1 = (f + g)(1) = f(1_{\mathbf{R}}) + g(1_{\mathbf{R}}) = 1 + 1 = 2$$

A contradiction! Hence $\text{Hom}(\mathbf{R}, \mathbf{S})$ is not a ring under pointwise addition and pointwise multiplication.

- (b)
 - (i) $\text{Hom}(\mathbb{Z}, \mathbb{Z}) = \{\text{id}\}$
 - (ii) $\text{Hom}(\mathbb{Z}/\langle 2 \rangle, \mathbb{Z}) = \emptyset$
 - (iii) $\text{Hom}(\mathbb{Z}, \mathbb{Z}/\langle 2 \rangle) = \{\mathbb{Z} \twoheadrightarrow \mathbb{Z}/\langle 2 \rangle\}$
 - (iv) $\text{Hom}(\mathbb{Z}, \mathbb{C}[x]) = \{\mathbb{Z} \hookrightarrow \mathbb{C}[x]\}$
- (c) Note that since both $\mathbb{Z}$ and $\mathbf{R}$ are rings, any homomorphism $\phi : \mathbb{Z} \rightarrow \mathbf{R}$ is unital, i.e. $\tau(1) = 1_{\mathbf{R}}$. To be a homomorphism we must have $\forall a \in \mathbb{Z}$,

$$\tau(a) = \tau\left(\sum_{i=1}^a 1\right) = \sum_{i=1}^a \tau(1)$$

This completely determines the homomorphism, and thus there is a unique (unital) homomorphism from $\mathbb{Z}$ to any ring $\mathbf{R}$.

Problem 4. Let $\phi : R \rightarrow S$ be a ring homomorphism.

- (a) Prove that if $J \subset S$ is an ideal then $\phi^{-1}(J) \subset R$ is an ideal.
- (b) Assuming ϕ is injective, give a nice description of $\phi^{-1}(J)$ for an ideal $J \subset S$.
- (c) Find a counterexample showing that if $I \subset R$ is an ideal $\phi(I)$ need not be an ideal.
- (d) Prove that if ϕ is surjective then $\phi(I)$ is an ideal of S .
- (e) Let $I \subset R$ be an ideal. Prove that if $\ker(\phi) \subset I$ then $\phi^{-1}(\phi(I)) = I$.
- (f) Is the assumption that $\ker(\phi) \subset I$ needed in part (5)?

Solution.

- (a) Let $a, b \in \phi^{-1}(J)$. Then $\phi(a), \phi(b) \in J$. Since ideals are closed under addition, $\phi(a) + \phi(b) = \phi(a + b) \in J$. Thus $\phi^{-1}(a + b) \in \phi^{-1}(J)$.
Similarly let $r \in R$, and $a \in \phi^{-1}(J)$. Then $\phi(r) \cdot \phi(a) = \phi(r \cdot a) \in J$. Thus, $\phi^{-1}(r \cdot a) \in \phi^{-1}(J)$.

- (b) If ϕ is injective then,

$$\phi^{-1}(J) = \{r \in R \mid \phi(r) \in J\}$$

$$\phi^{-1}(J) = J \cap \phi(R) = J \cap R$$

- (c) Let $R = \mathbb{Z}$, $S = \mathbb{Z}[x]$ and define the homomorphism to be $\phi : R \hookrightarrow S$. Then consider the image of the ideal generated by 3. It does not contain $3x$ and hence is not closed under multiplication.

- (d) Let $I \subset R$ be an ideal and assume that ϕ is surjective. Let $a, b \in I$. We want to show that $\phi(I)$ is a subgroup of the additive group of S and is closed under multiplication by elements of S .

Consider $\phi(a) - \phi(b)$. Note that since ϕ is a homomorphism the following holds,

$$\phi(a) - \phi(b) = \phi(a - b)$$

and since $a - b \in I$, $\phi(a - b) \in \phi(I)$. Moreover since $\phi(0) = 0 \in \phi(I)$, $\phi(I)$ is a subgroup of S under addition.

Now note that since ϕ is surjective, $\forall s \in S \exists r \in R$ such that $s = \phi(r)$. Then since $a \in I$, $r \in R$ and I is an ideal of R we have $ra \in I$. So,

$$s \cdot \phi(a) = \phi(r)\phi(a) = \phi(ra) \in \phi(I)$$

Hence, $\phi(I)$ is closed under multiplication by elements of S and thus is an ideal of S .

(e) Assume $\ker(\phi) \subset I$.

$$I \subset \phi^{-1}(\phi(I))$$

Since $\phi^{-1}(\phi(I))$ is the pre-image of $\phi(I)$, this containment is straightforward.

$$\phi^{-1}(\phi(I)) \subseteq I$$

Let $x \in \phi^{-1}(\phi(I))$. That is $\phi(x) = \phi(a)$ for some $a \in I$.

$$\phi(x - a) = 0 \implies x - a \in \ker(\phi) \subset I$$

Since I is closed under addition $x \in I$. Hence, $\phi^{-1}(\phi(I)) \subseteq I$

(f) Yes. As a counterexample, consider the quotient map $\phi : \mathbb{Z} \rightarrow \mathbb{Z}/\langle 3 \rangle$ and the ideal generated by 2, $I = \langle 2 \rangle$. Note that $0 \in \phi(I) \implies \ker(\phi) = 3\mathbb{Z} \subset \phi^{-1}(\phi(I))$. That is $3 \in \phi^{-1}(\phi(I))$ but $3 \notin I$.